

flip the object in a different orientation? This will be an entirely new set of 30 different technologies! However, the customers who utilise our tech don't have to change the technology structure or the product structure.

What kind of electronics is used in your products?

Gokul: Just like an iMac has a processor from Intel and a screen from some other vendor, only a portion of our systems are made entirely in-house. For most commercial products, it's the same case. Since we are at a seed stage, our core tech—vision—is the hardware stack that is completely built in-house.

For the customers, we bring in a lot of other technologies—the robotic arm, the gripper, some software layers, the processing elements that are to be custom designed for the industrial scenario, etc. We provide our customers with the whole system—a robotic arm equipped with vision and with sophisticated grasping capabilities, so that it can handle different kinds of objects.

Nikhil: It's a sophisticated, human-like camera setup. Many of our parts are custom-designed for us. Our components come from all over the world. So, typically, the first version is an amalgamation of technology that originates across the world and comes into that one small focal point of the camera that we design. Eventually, we'll have manufacturing operations to streamline the production.

We've already launched our next iteration, where we will be making it a lot more compact. We'll keep having newer versions of the camera where we either improve the specs, reduce the size and dimensions, or build versions that can allow us to scale to newer application use cases.

What has been your commercialisation story?

Nikhil: We are actually com-



CynLr's Visual Robot

mercialising as we speak. The good thing is that for what we're building, the customer and market requirements are clearly established. A robot that has the intelligence to pick an object and do something with it has been a dream for almost 40 or 50 years, ever since the first robotic arm was built.

From the customer's point of view, and a commercial standpoint, we've usually found very little resistance. The challenge is on the supply side of it because there are a lot of parts that come from all over the world. So, that's something that we are actively working to iron out in this space.

We are setting up a sophisticated supply-chain process, and we already have an ERP system in place. We're also investing in SAP-like tools, so that we can manage all the supply-chain issues in manufacturing and delivery. We already have a pipeline of customers and an OEM partner. The road to commercialisation is underway.

What kind of challenges have you faced so far?

Gokul: On the technology side, the lack of awareness and narrative are challenging. On any media front, the term 'machine vision' is uncommon. For example, on LinkedIn, #machinevision will not have as many followers as #computervision. So, that is the perspective awareness variation, and this influences the supply-chain too. Also, the amount of time that you spend explaining that foundational concept increases, right?

People often get confused between technology and application. Applications are easier to articulate because you're building on top of a technology that is well established. You don't have to explain what the internet is, because people already know what the internet is! You don't have to explain what an Android phone is, because people already have an Android phone in their hands. But imagine a company like Xilinx or Nvidia trying to